LAST NAME: _____

Practice Quiz 10C — MATH 2055 — Fall 2026

Practice for Quiz 10

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. Let $a_1 = a_2 = 1$, and define

$$a_n = \frac{a_{n-1} + a_{n-2}}{3} \qquad (n \geq 3).$$

Prove by strong induction that $0 < a_n \leq 1$ for every integer $n \geq 1$. State the complete inductive hypothesis and check both initial cases.